Big mart sales



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Abstract

This project focuses on analyzing and predicting sales performance in retail outlets by examining how various product attributes, such as category, pricing, visibility, weight, and store characteristics affect sales. The dataset, containing 8,523 records and 12 features, revealed key insights that can guide retailers in optimizing their strategies.

Products like FDY55, FDA15, FDZ20, and FDA04 were top performers in sales, suggesting that marketing and promotional efforts should focus on these items. In contrast, FDJ55 and FDD44 showed lower sales among the top 10. Contrary to expectations, product display space and weight had minimal impact on sales DRE01 and FDS12 occupied the largest shelf space but did not translate into higher sales. Interestingly, low-fat products outperformed regular ones in terms of sales, yet regular products yielded higher profits.

Supermarket Type 1 led in overall sales, followed by grocery stores, while fruits and vegetables were the most profitable categories, followed by snack foods. On the other hand, breakfast products and seafood lagged behind, potentially due to customer preferences or product quality. Frozen foods, canned goods, and baking products were widely stocked across all outlets, emphasizing their importance in inventory.

In terms of store tiers, Tier 2 stores achieved the highest average sales percentage despite having fewer stores than Tier 3, which had the widest product variety. Mid-sized stores outperformed other store sizes in average sales, and Supermarket Type 3 achieved the highest sales overall, while grocery stores ranked lowest.

In addition to these analyses, a machine learning model was developed to predict Item Outlet Sales. Before building the model, extensive data preprocessing steps were carried out:

1. Handling missing values,
2. Removing redundant features,
3. Addressing feature correlations,
4. Managing outliers,
5. Correcting skewness in the data,
6. Ensuring features followed a normal distribution,
7. Splitting the data into training and testing sets,
8. Encoding categorical features,
9. Normalizing the data,
10. Handling multicollinearity.

Introduction

In the highly competitive world of retail, understanding the factors that influence product sales is crucial for optimizing business strategies. Retailers are constantly seeking ways to maximize their revenue by improving inventory management, product placement, and promotional tactics. However, the complexity of sales data—ranging from product attributes like weight and fat content to store characteristics such as size and type—presents a challenge when it comes to making data-driven decisions.

This project tackles the problem of predicting Item Outlet Sales by analyzing various product and store attributes. By leveraging machine learning techniques, we aim to uncover the key factors driving sales performance across different retail outlets. Identifying these factors can help retailers fine-tune their strategies, whether by focusing on high-performing products, adjusting pricing models, or optimizing stock levels for specific store types.

Through a detailed data analysis and the application of advanced machine learning algorithms, this project offers insights into how different elements, such as product visibility, pricing, and store characteristics, contribute to overall sales. The ultimate goal is to build a predictive model that enables retailers to forecast sales more accurately and make informed decisions that enhance profitability and customer satisfaction.

By utilizing data from over 8,500 records, the project not only highlights current sales trends but also provides actionable recommendations for optimizing operations. The results are geared towards benefiting retail managers, marketing teams, and product managers, empowering them to turn insights into effective business strategies.

This report will cover the problem, methodology, model performance, and results, demonstrating how machine learning can transform the way retailers approach sales forecasting and decision-making in an increasingly data-driven world.

Various machine learning models were tested for prediction accuracy, summarized as follows:

Mean Squared Error (MSE):

- **Highest MSE:** XGBRegressor (954.408), GradientBoostingRegressor (953.297)
- **Lowest MSE:** Ridge (1021.112)
- **Model Ranking (MSE):** XGBRegressor > GradientBoostingRegressor > Random Forest (965.210) > Decision Tree (953.367) > SVR (1041.309) > Ridge

2. Mean Absolute Error (MAE):

- **Highest MAE:** XGBRegressor (724.08), GradientBoostingRegressor (721.51)
- **Lowest MAE:** Ridge (770.53)
- **Model Ranking (MAE):** Similar to MSE, with XGBRegressor and GradientBoostingRegressor having the highest values, followed by Random Forest, Decision Tree, SVR, and Ridge.

3. R^2 Score:

- **Best R^2 Scores:** XGBRegressor (0.6188), GradientBoostingRegressor (0.6193), Random Forest (0.6145)
- **Lowest R^2 Scores:** Ridge (0.5922), SVR (0.5841)
- **Model Ranking (R^2):** XGBRegressor ≈ GradientBoostingRegressor > Random Forest > Decision Tree > SVR > Ridge

Model Performance Summary:

- **Best overall:** Random Forest and GradientBoostingRegressor showed the most balanced performance in terms of both accuracy and generalization.
- **Competitive:** XGBRegressor performed competitively and could be further optimized.
- **Underperformers:** Ridge and SVR underperformed compared to the more complex models.

*Random Forest and **GradientBoostingRegressor** delivered the best overall performance with balanced accuracy and generalization. **XGBRegressor** also showed competitive results and could be further optimized. **Ridge Regression** and **SVR** underperformed compared to the more complex models.*

*Finally, a **Voting Regressor** was created by combining these models, achieving the following results:*

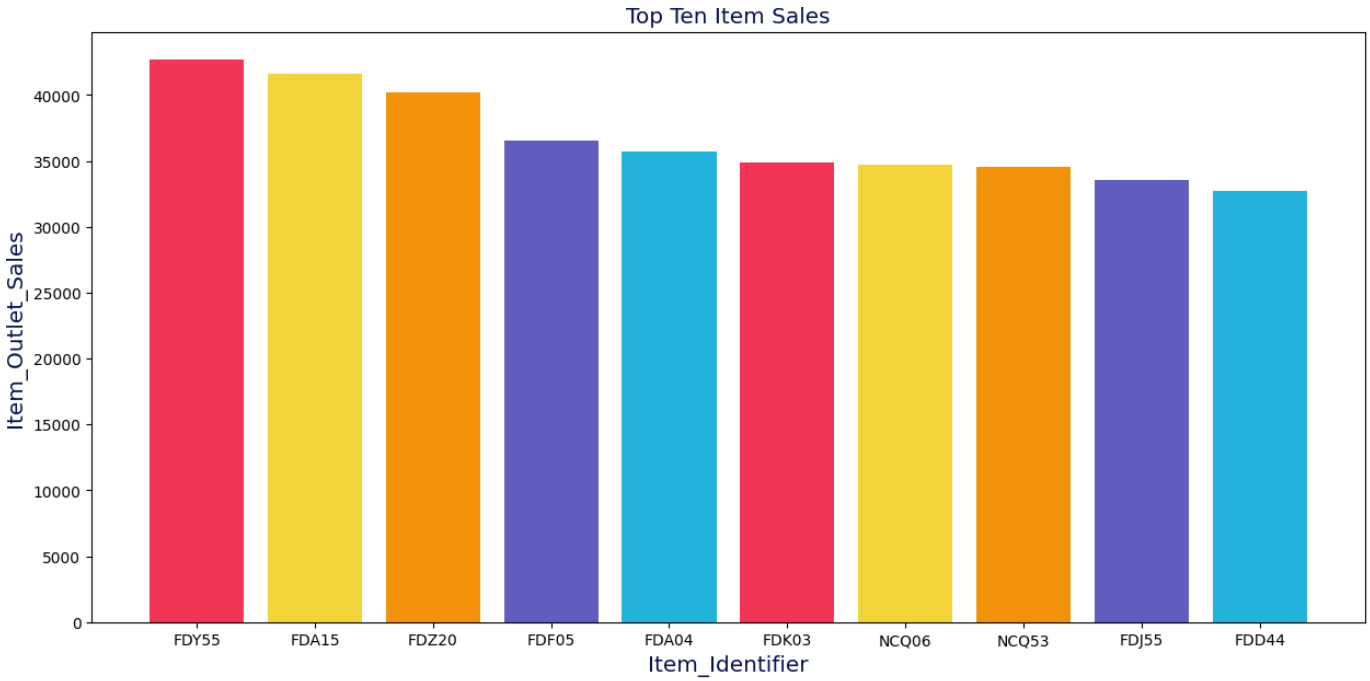
Data	*MSE*	*MAE*	*R2*
Train	987.120	735.59	0.6306
Test	946.632	717.66	0.6220

These results represent the best performance achievable based on this dataset, offering a strong predictive model for Item Outlet Sales. This model can aid retailers in making data driven decisions for optimizing product placements, pricing strategies, and inventory management.

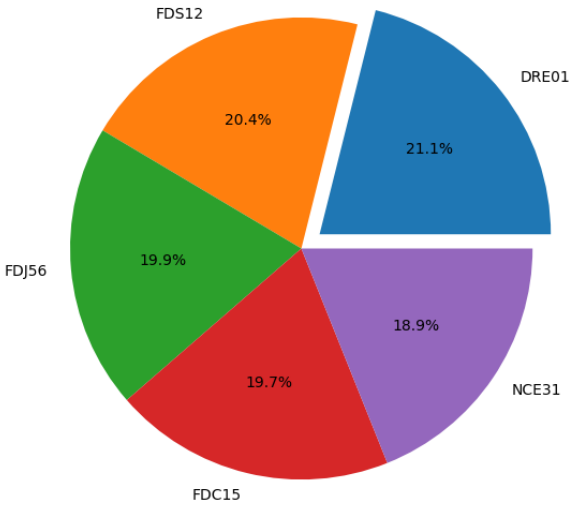
Methods:

1. **Data Analysis and Understanding**:
   - Analyze the dataset to gain insights and understand the key characteristics.
   - Identify the features that have the most significant impact on the target variable.
2. **Feature Selection**:
   - Determine the most important features influencing the target.
3. **Apply Machine Learning Algorithms**:
   - Implement various machine learning algorithms on the dataset to predict the target variable.
   - Evaluate the performance of different models to find the best one.

• Data Analysis



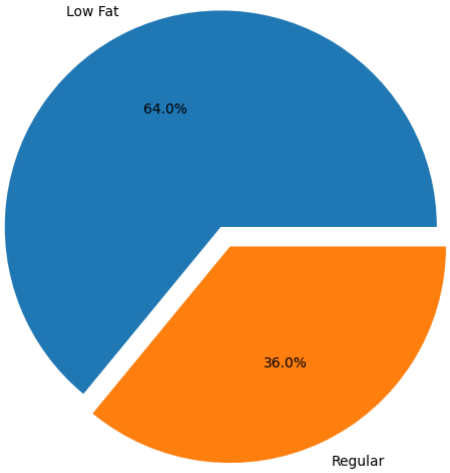
- *FDY55 and FDA15 are the highest-performing products in terms of sales.*
- *FDZ20 and FDA04 also show strong sales, though slightly lower than the top two.*
- *Products like FDJ55 and FDD44 have relatively lower sales, suggesting reduced demand or popularity compared to other products.*



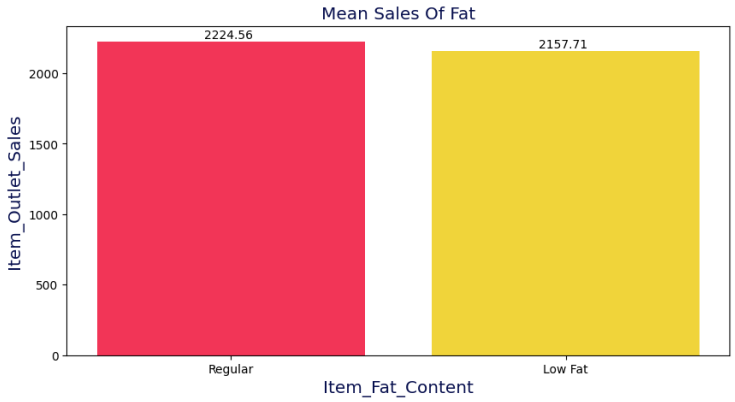
*it is clear that there is not much difference in **Item_Visibility** across different types of products (Item_Identifier). Some products, such as DRE01 and FDS12, have higher Visibility, while others like FDJ56 , FDC15 and NCE31 have relatively lower Visibility.*

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*Based on the analysis of the two graphs, it is clear that products with higher **Visibility** are not necessarily the ones with the highest sales, and vice versa. This suggests that other factors, such as product type or price, may play a larger role in driving sales.*



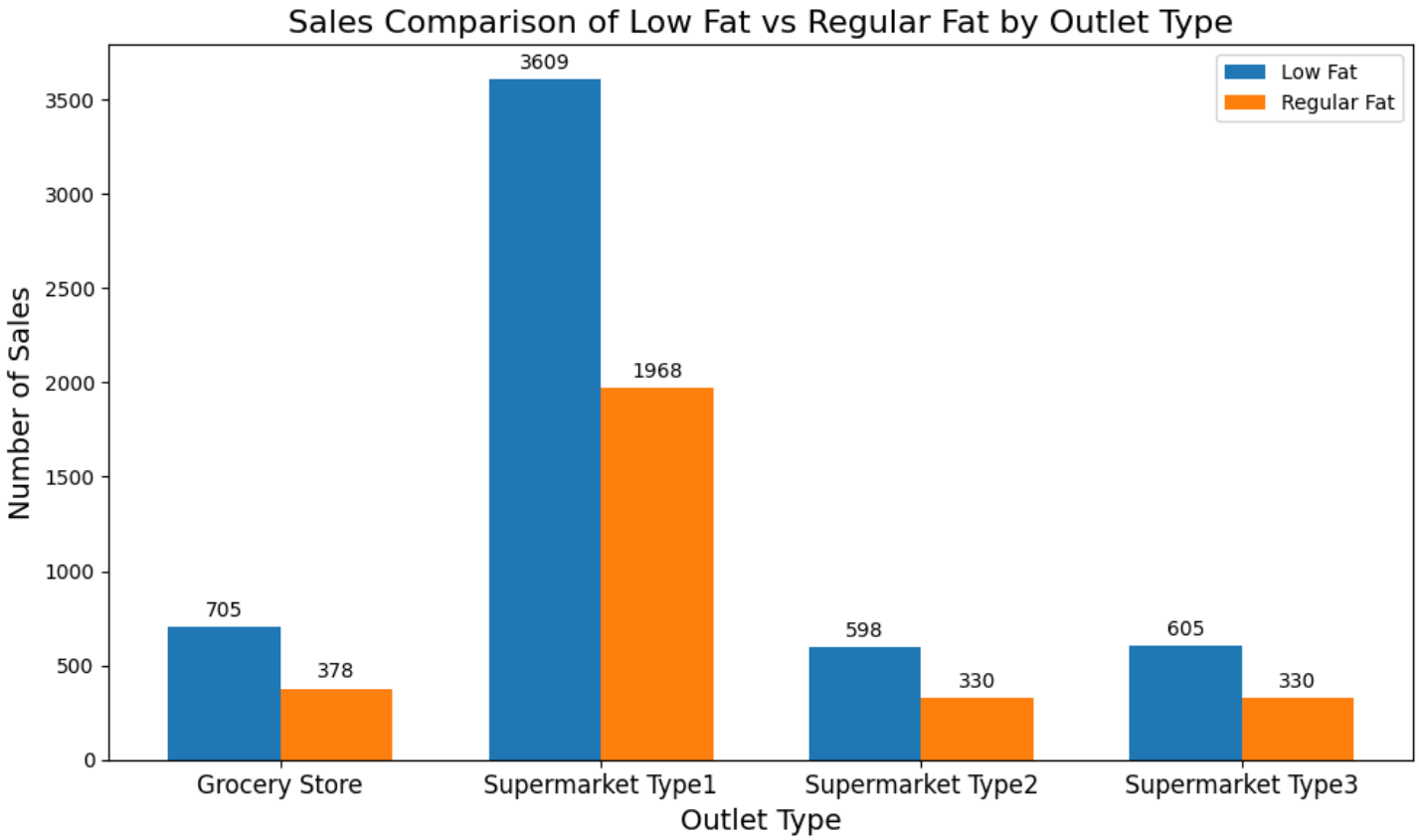
Across all outlet



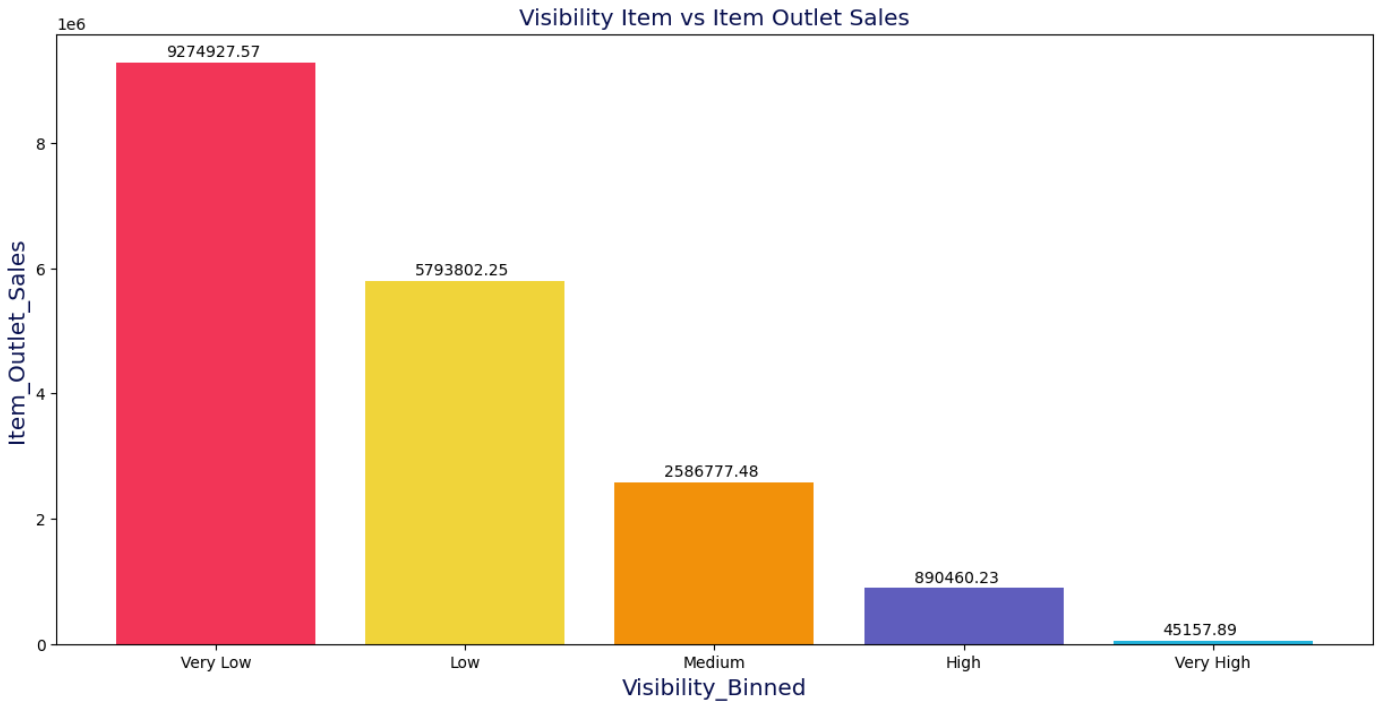
*types, **Low Fat** items consistently outsell **Regular fat** items.*

*it is clear that items with **low fat** content have higher sales compared to those with **regular fat** content*

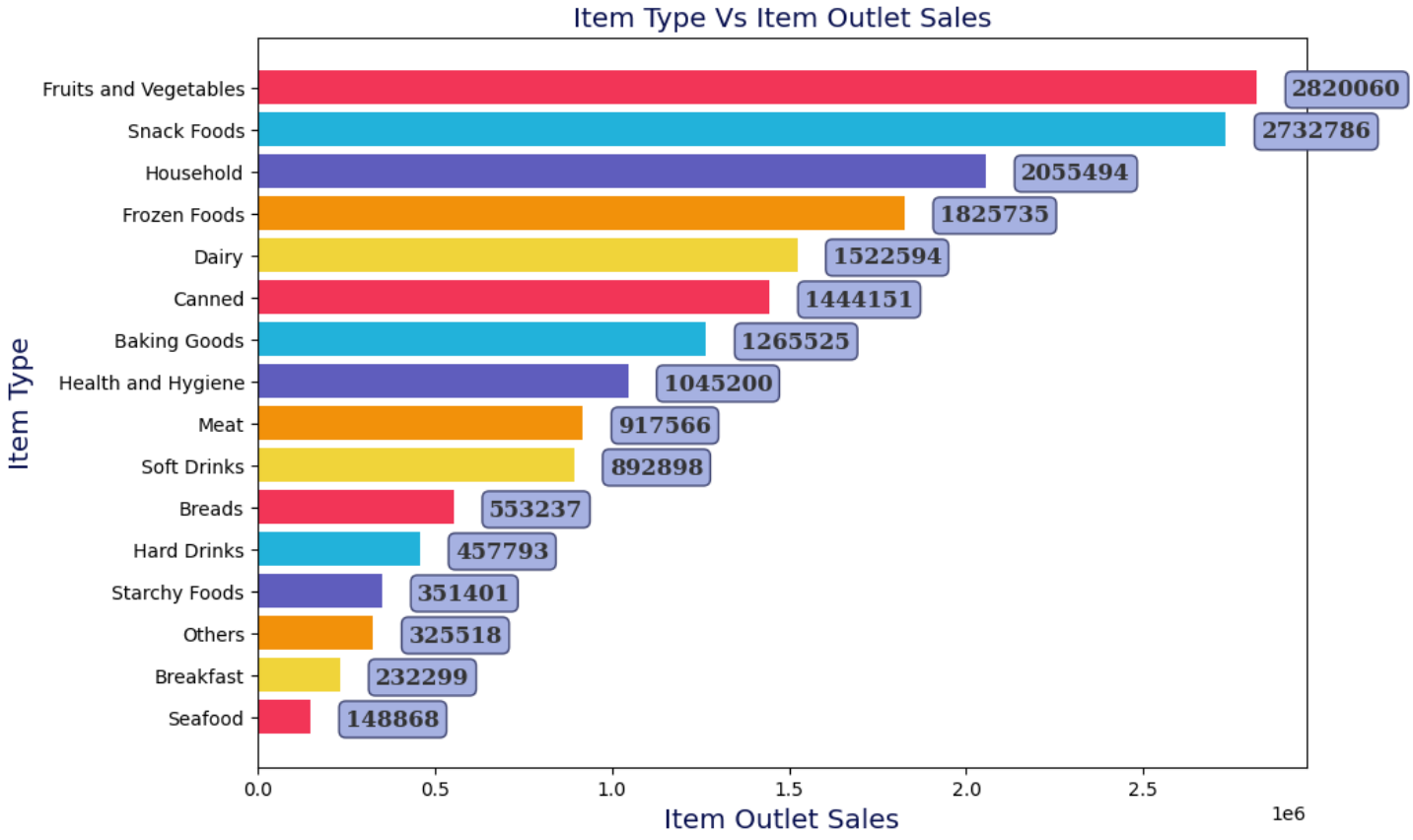
*Although **Low Fat** items generate higher sales in terms of quantity, **Regular Fat** items yield higher profits.*



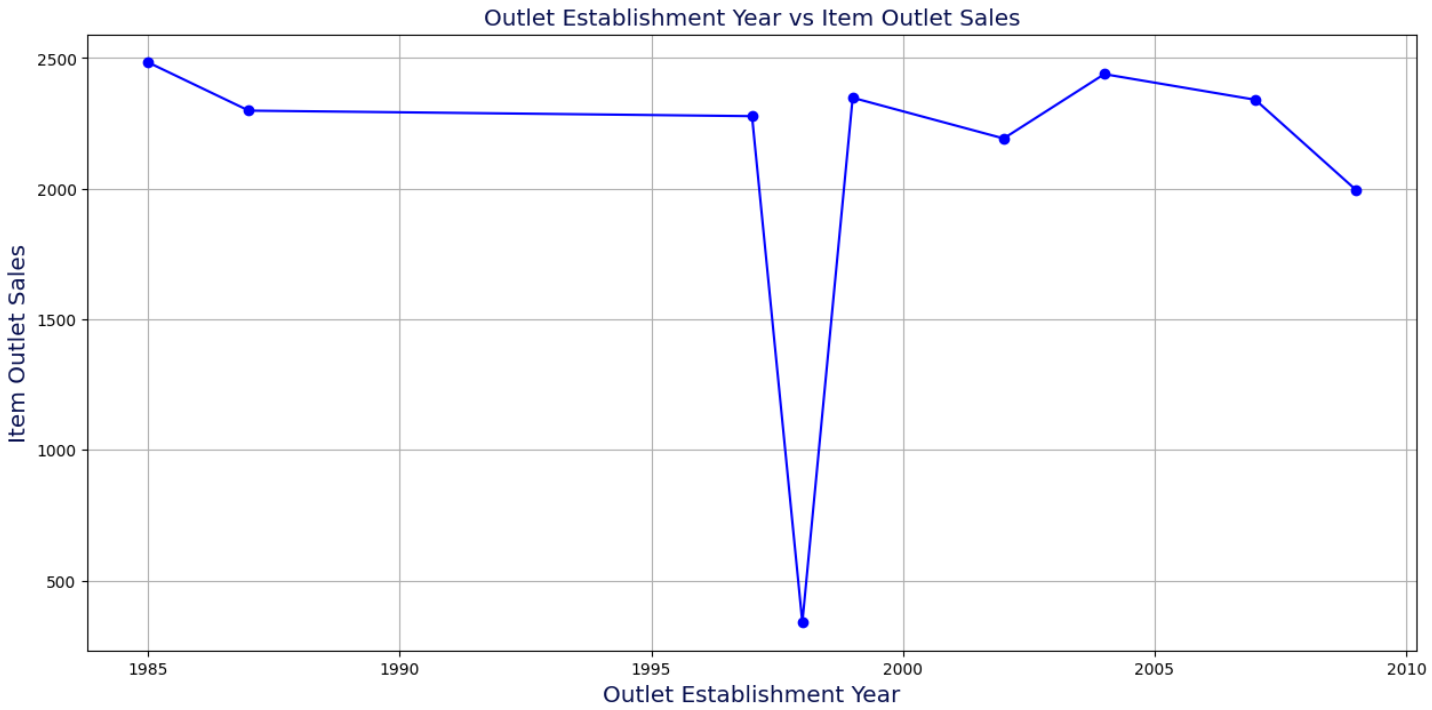
- *Supermarket Type1 has the highest sales volume for both **Low Fat** and **Regular** items, indicating it may have a larger customer base or better product variety.*
- *Grocery Stores and Supermarket Type2/Type3 show lower overall sales compared to Type1 supermarkets, but the trend of **Low Fat** items being more popular is consistent.*



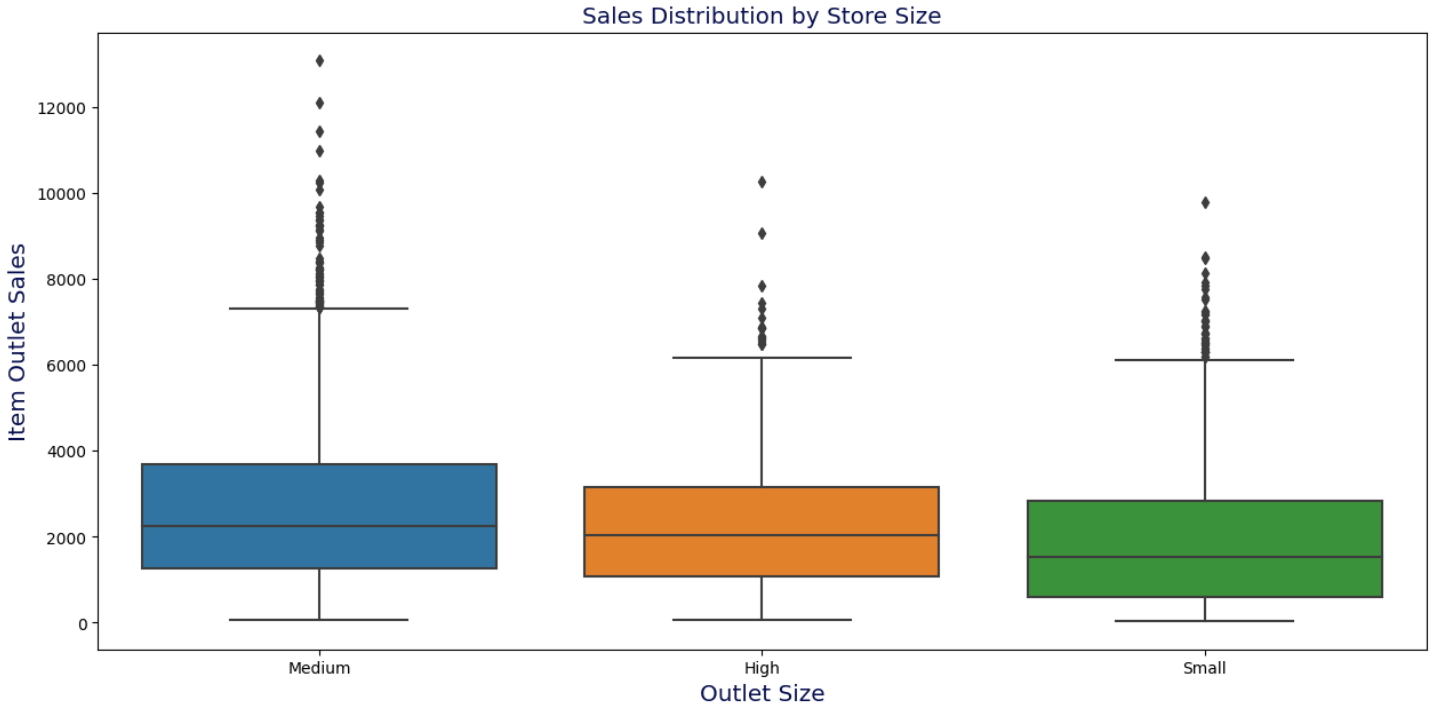
It is very clear from the graph that the product with the lowest visibility size is the most profitable.



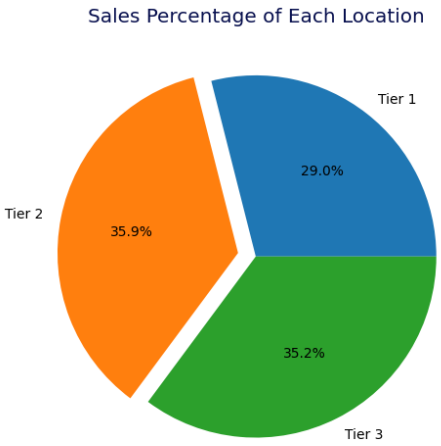
- *It is evident that **fruits and vegetables** are the top-performing categories in terms of sales, followed closely by **snack foods**.*
- *On the other hand, **seafood** and **breakfast** items are among the lowest-performing categories in terms of sales.*

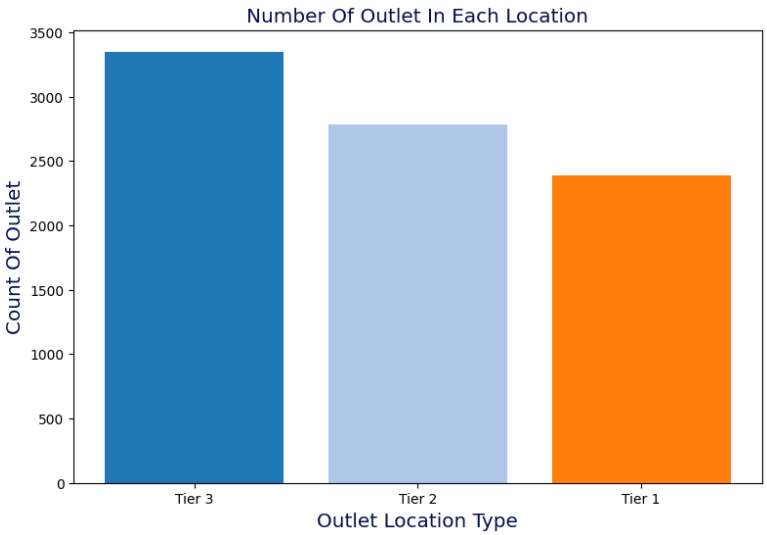


- *1985: The average sales were the highest compared to all other years.*
- *1986 to 1997: The average sales remained relatively stable during this period.*
- *1998: A significant decline in sales occurred in this year.*
- *1999: Sales began to recover and showed an upward trend.*
- *2003: The outlets achieved a very high average sales, comparable to the sales in 1985.*
- *2004 to 2009: There was a gradual decline in the average sales, leading to a decrease by 2009.*

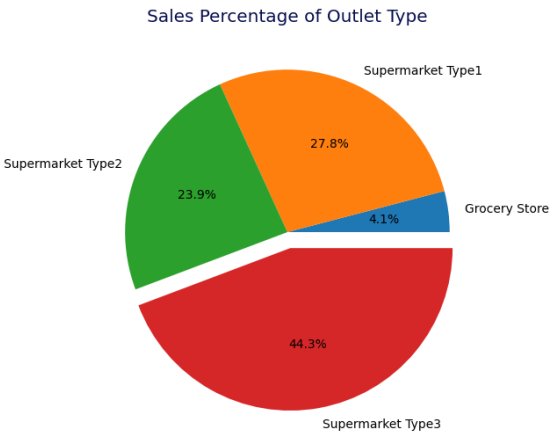


We conclude that medium-sized stores have higher average sales compared to the other store sizes. However, the differences are not very large. Still, medium-sized stores show sales that appear significantly higher when compared to both large and small stores, with these large and small stores acting as outliers.





- *The stores in Tier 2 have the highest average sales percentage.*
- *The number of stores in Tier 3 is the highest. However, despite this, Tier 2, with its moderate number of stores, has the highest percentage of profit.*



Supermarket Type 3 achieves the highest average sales, while grocery stores have the lowest sales percentage.

The Conclusion of Analysis

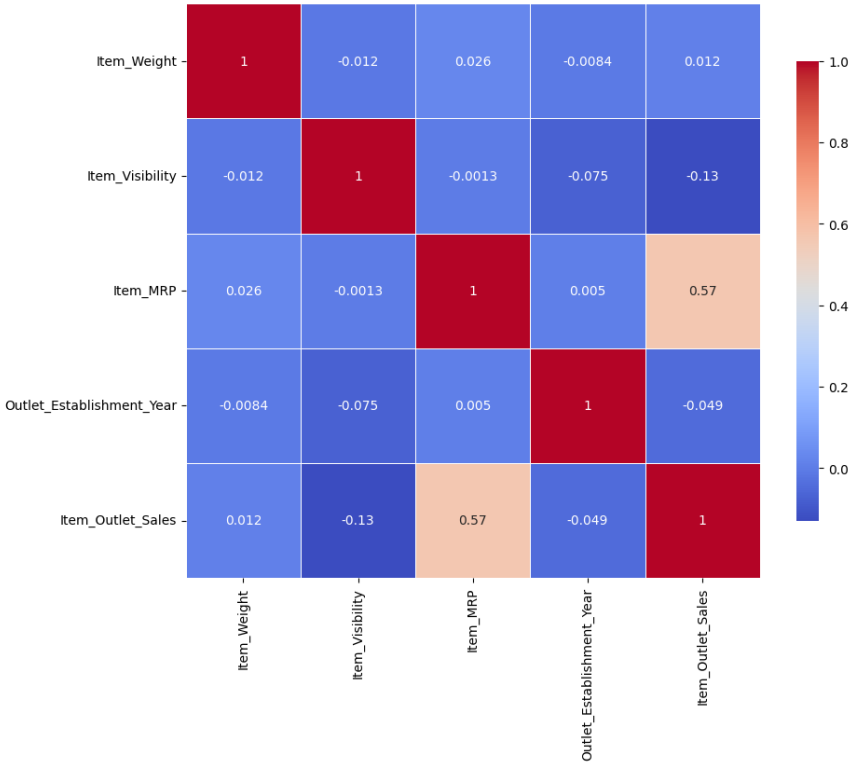
This analysis of sales performance across different product categories and outlets has provided valuable insights into factors that significantly influence sales. The findings indicate that while certain products such as FDY55 and FDA15 show the highest sales figures, visibility alone does not correlate with sales. Other factors such as product type, fat content, and price are found to play a more significant role in driving sales performance.

Key Findings:

- FDY55 and FDA15 are top sellers, while FDJ55 and FDD44 lag.
- No direct link between visibility and sales; product type and price matter more.
- Low-fat products sell more, but regular-fat ones are more profitable.
- Type 1 supermarkets outperform smaller outlets, especially for low-fat items.
- Fruits, vegetables, and snacks perform best; seafood and breakfast lag.
- Medium-sized stores show higher sales than large and small ones.
- Tier 2 stores have the highest profit margins despite fewer outlets.

- # Machine Learning

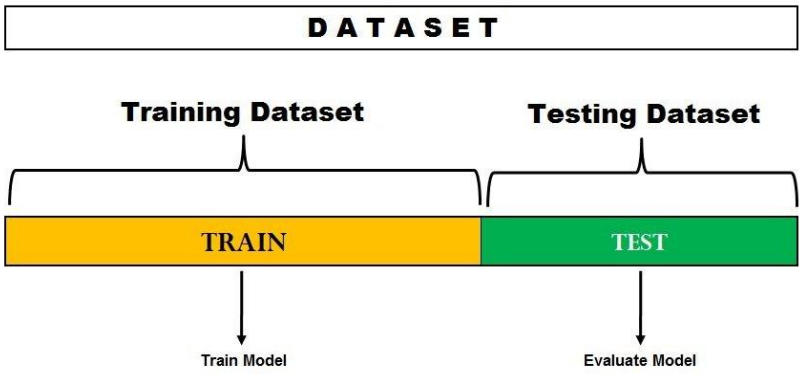
- *Correlation Coefficient*



1. **Item MRP and Item Outlet Sales**: There is a moderately strong positive correlation (0.57), indicating that as the maximum retail price (MRP) of items increases, the outlet sales tend to increase as well.
2. **Item Visibility and Item Outlet Sales**: There is a negative correlation (-0.13), suggesting that items with higher visibility are not necessarily generating higher sales, which may indicate that visibility alone is not a strong driver of sales.
3. **Item Weight**: The correlations with other variables, including sales, are very low, suggesting that item weight does not play a significant role in determining sales or other factors like visibility and MRP.
4. **Outlet Establishment Year**: The correlation with Item Outlet Sales is weak (-0.049), meaning the age of the outlet doesn't significantly affect sales performance.

Overall, the strongest relationship is between **Item MRP and Item Outlet Sales**, highlighting that price is an important factor in driving sales in this dataset. Other factors, like visibility and weight, show weaker or insignificant correlations.

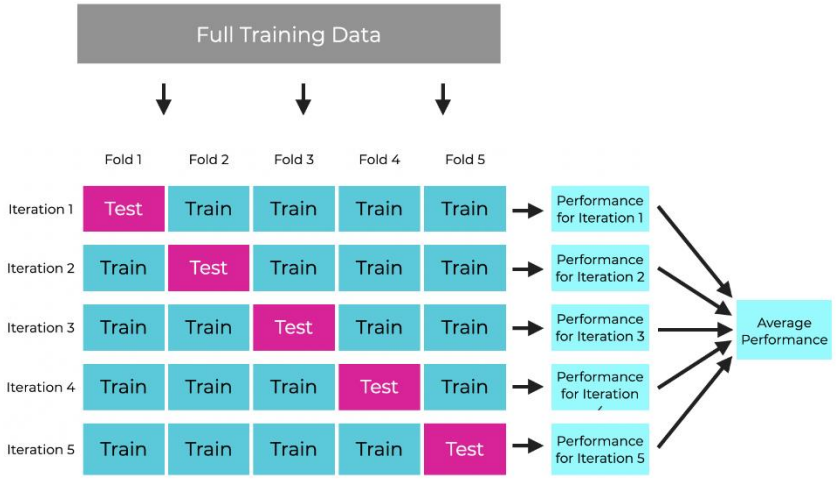
- *Data Spliting*



The dataset was split into two parts:

- **80% for training**: This portion of the data is used to train the machine learning models. The models learn patterns, relationships, and dependencies within the data to make predictions.
- **20% for testing**: This portion is kept aside to evaluate the performance of the trained models. It allows us to assess how well the model generalizes to unseen data, ensuring the model isn't overfitted to the training set.

- *Cross Validation*



Cross-validation is a technique used to evaluate a model by splitting the data into multiple parts (or "folds"). The model is trained on some of these parts and tested on the remaining part, and this process is repeated several times to ensure the model performs well on different subsets of the data. This helps in avoiding overfitting and gives a more accurate assessment of the model's performance.

In our case, we applied cross-validation internally with `RandomizedSearchCV`. As we searched for the best hyperparameters, the data was split into 3 parts (using `cv=3`), where the model was trained on 2 parts and tested on the remaining part. This process was repeated 3 times for each random combination of hyperparameters, making sure the model performs well across different data splits.

- *Handle Categorical Features*

The categorical features were encoded as follows:

1. **Ordinal Encoding**:
    - Applied **Label Encoding** on the ordinal categorical feature `Item_Type`, which implies a meaningful order or ranking in the values. The label encoder assigns integer values to each category based on the inherent order.
2. **Nominal Encoding**:
    - Applied **OneHot Encoding** on the nominal categorical features: `Item_Fat_Content`, `Outlet_Size`, `Outlet_Location_Type`, `Outlet_Type`, and `Visibility_Binned`. Since these features do not have an inherent order, OneHot Encoding was used to create binary columns for each category, ensuring the model doesn't assume any ranking between the categories.

- *Feature Scaling*

The **MinMaxScaler** was applied to all numerical columns to scale their values within a specific range (usually between 0 and 1). This ensures that the numerical features are on the same scale, improving the performance of models like **Support Vector Machines**, **K-Nearest Neighbors**, and other algorithms sensitive to varying data ranges.

- *Build Models*

We evaluate the performance of multiple regression models without applying any hyperparameter tuning. The following models were selected for comparison:

1. Linear Regression, 2. Lasso Regression, 3. Ridge Regression, 4. ElasticNet Regression, 5.(KNN),
6. Support Vector Regression (SVR), 7. Decision Tree Regressor, 8. Random Forest Regressor,
9. AdaBoost Regressor, 10. Gradient Boosting Regressor, 11. XGBoost Regressor

For each model, we used the default hyperparameters (without any optimization) and trained them on the training dataset. After training, the models were evaluated on the test dataset using key performance metrics: **Mean Squared Error (MSE)**, **Mean Absolute Error (MAE)**, and **R-Squared (R^2)**. This evaluation will provide a baseline comparison of model performance before any further optimization or tuning.

- **GradientBoostingRegressor** demonstrates the best overall performance, showing an R^2 of **0.617** on the test set, with balanced **MAE** and **MSE** values.
- **XGBRegressor** follows closely, with an R^2 of **0.540** on the test set, showcasing strong generalization abilities with competitive errors.
- Models such as **Decision Tree** and **Random Forest** exhibit signs of overfitting, performing excellently on the training set but much worse on the test set.
- **SVR** and **ElasticNet** underperform, with negative or very low R^2 values, suggesting they struggle to fit the data appropriately.

GradientBoostingRegressor appears to be the optimal model, offering the best balance between predictive accuracy and generalization. However, with further tuning, **XGBRegressor** may also yield strong results.

- *Hyperparameters tuning*

I used RandomizedSearchCV and applied cross-validation by splitting the data into 3 parts. The results of the tuning were as follows:

- **Ridge Regression**: Decent performance with moderate MAE, RMSE, and an R^2 of 0.59.
- **SVR**: Weaker performance with an R^2 of 0.58 and higher MAE and MSE compared to Ridge.
- **Decision Tree**: Solid generalization with an R^2 of 0.62 but slight overfitting.
- **Random Forest**: Strong performance with an R^2 of 0.61, balancing bias and variance well, though with slight overfitting.
- **GradientBoostingRegressor**: Excellent performance with an R^2 of 0.62, similar to Random Forest.
- **XGBRegressor**: Performs similarly to GradientBoostingRegressor and Random Forest with an R^2 of 0.62.

Overall, Random Forest and GradientBoostingRegressor lead the performance with well-balanced accuracy and generalization. However, XGBRegressor shows competitive results and could be further optimized. Ridge Regression and SVR underperform compared to the more complex models.

- # Result and Evaluation

- ## *Final Model*

For the final model, we implemented a **Voting Regressor**, which combines the predictions from multiple individual regressors to provide a robust, ensemble-based solution. The Voting Regressor aggregates the predictions of several models, which typically leads to improved performance by leveraging the strengths of each individual model.

The following models were included in the ensemble:

1. **Ridge Regression** (with 'lsqr' solver and regularization strength of 1.0)
2. **Support Vector Regression (SVR)** (with linear kernel and regularization parameter C=100.0C=100.0C=100.0)
3. **Decision Tree Regressor** (with specific settings for tree depth and splitting criteria)
4. **Random Forest Regressor** (with 100 trees and specific splitting criteria)
5. **Gradient Boosting Regressor** (with 50 estimators and moderate learning rate)
6. **XGBoost Regressor** (with 50 estimators and moderate learning rate)

The **Voting Regressor** was trained using the training dataset and evaluated on the validation dataset. The model aggregates the outputs of the individual regressors and computes a final prediction. The **R^2 score** for the Voting Regressor on the validation data was calculated to be **[insert R^2 score]**, indicating its overall performance.

This ensemble approach was chosen for its ability to minimize the risk of overfitting and to combine the complementary strengths of different regression models.

Data	*MSE*	*MAE*	*R2*
Train	*987.120*	*735.59*	*0.6306*
Test	*946.632*	*717.66*	*0.6220*

Overall, the final model performs well, with good predictive accuracy as indicated by the R^2 values. However, there is a slight performance gap between the training and validation sets, which could be further minimized with additional tuning or regularization techniques. The model demonstrates solid generalization, and further improvements could be made through data augmentation or advanced hyperparameter tuning.

Conclusion

- Recommendations to Increase Sales:
  - Focus marketing and promotional efforts on top-selling products like FDY55 and FDA15. Highlight these products in advertisements and store displays to attract customer attention.
  - Ensure that high-demand items, especially fruits, vegetables, and low-fat products, are consistently in stock. Monitor sales trends to adjust inventory levels dynamically and avoid stockouts.
  - While visibility alone does not guarantee higher sales, improving the placement of high-performing products in stores can help. Use eye-catching displays and strategic shelf placement to draw attention to these items.
  - Consider competitive pricing strategies for high-demand products. Analyze pricing of similar products in the market and adjust prices to attract price-sensitive customers while ensuring profitability.
  - Introduce new variations or complementary products in categories that show high demand, such as frozen foods and baking goods. This can encourage cross-selling and increase overall sales.
  - Given that low-fat products are performing better in sales, consider launching marketing campaigns that promote the health benefits of these items. This can attract health-conscious consumers.

After applying hyperparameter tuning to the different models, a noticeable improvement in performance was observed, with the models performing well on both the **training** and **test** datasets. By adjusting various parameters, the accuracy of the predictions was enhanced, and the gap between performance on training and test data was minimized, effectively avoiding **overfitting** and achieving more stable results.

To further boost the models' performance and unify their outputs, an **Ensemble** technique was applied through the **Voting Regressor**. This ensemble method combines multiple models and merges their predictions to yield a more accurate forecast. The **Voting Regressor** uses majority voting from individual model predictions (such as linear regression, random forests, etc.) to create a unified prediction that reflects the strength and reliability of each individual model.

Through this approach, the system was able to leverage the strengths of each model, leading to precise results on both **training** and **test** datasets. This technique improved stability and reduced variability in predictions, allowing the system to handle a wide range of scenarios effectively.